

Limiti notevoli

$x \rightarrow 0$

- $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\arcsin(x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\tan(x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\arctan(x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\cos(x)}{x^2} = \frac{1}{2}$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$

$x \rightarrow \infty$

- $\lim_{x \rightarrow +\infty} (1 + \frac{\alpha}{x})^x = e^\alpha$
- $\lim_{x \rightarrow +\infty} x^{\frac{1}{x}} = 1$
- $\lim_{x \rightarrow +\infty} P(x)^{\frac{1}{x}} = 1$

Gerarchia infiniti

$$(\log_B n)^\alpha < n^\alpha < A^n < n! < n^n$$

Derivate

- $[x^\alpha]' = \alpha x^{\alpha-1}$
- $[e^x]' = e^x$
- $[\sin(x)]' = \cos(x)$
- $[\cos(x)]' = -\sin(x)$
- $[\tan(x)]' = \frac{1}{\cos^2(x)}$
- $[\arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$
- $[\arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$
- $[\arctan(x)]' = \frac{1}{1+x^2}$

Regole di derivazione

- $[kf(x)]' = kf'(x)$
- $[f(x) \pm g(x)]' = f'(x) \pm g'(x)$
- $[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$
- $[\frac{f(x)}{g(x)}]' = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$
- $[f(g(x))]' = f'(g(x))g'(x)$

Integrali

- $\int_a^b 1 dx = [x]_a^b$
- $\int_a^b x^n dx = [\frac{x^{n+1}}{n+1}]_a^b$
- $\int_a^b \sin x dx = [-\cos x]_a^b$
- $\int_a^b \cos x dx = [\sin x]_a^b$
- $\int_a^b \frac{1}{\cos^2 x} dx = [\tan x]_a^b$
- $\int_a^b \frac{1}{\sin^2 x} dx = [-\frac{1}{\tan x}]_a^b$
- $\int_a^b \frac{1}{x^2} dx = [\arctan x]_a^b$
- $\int_a^b \frac{1}{x} dx = [\ln |x|]_a^b$
- $\int_a^b e^x dx = [e^x]_a^b$

Regole di integrazione

- $\int_a^b k \cdot f(x) dx = k \int_a^b f(x) dx$
- $\int_a^b f(x) \pm g(x) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$

Media integrale

$$\langle f \rangle = \frac{1}{b-a} \int_a^b f(x) dx$$

Serie numeriche

Serie geometrica

$$\sum_{n=p}^{+\infty} x^n = x^p \sum_{i=0}^{+\infty} x^i = \begin{cases} +\infty & \text{se } x > 1 \\ x^p \frac{1}{1-x} & \text{se } -1 < x < 1 \\ \varnothing & \text{se } x \leq -1 \end{cases}$$

Serie armoniche

$$\sum_{n=0}^{+\infty} \frac{1}{n^\alpha} : \begin{cases} \text{Diverge} & \text{se } \alpha \leq 1 \\ \text{Converge} & \text{se } \alpha > 1 \end{cases}$$

Serie telescopiche

Sia a_n una serie tale che $a_n = b_{n+1} - b_n$. Si ha

$$S_n = \sum_{k=1}^n a_k = b_{n+1} - b_1$$

Quindi

$$\lim_{n \rightarrow +\infty} S_n = \sum_{n=1}^{+\infty} a_n = \lim_{n \rightarrow +\infty} b_{n+1} - b_1$$

Criteri

- Radice

$$\sum_{n=0}^{+\infty} a_n : \begin{cases} \text{Diverge} & \lim_{n \rightarrow +\infty} \sqrt[n]{a_n} > 1 \\ \text{Converge} & \lim_{n \rightarrow +\infty} \sqrt[n]{a_n} < 1 \end{cases}$$

- Rapporto

$$\sum_{n=0}^{+\infty} a_n : \begin{cases} \text{Diverge} & \lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} > 1 \\ \text{Converge} & \lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} < 1 \end{cases}$$

- Leibnitz

$$\text{Se } \sum_{n=0}^{+\infty} (-1)^n a_n : \begin{cases} a_n \geq 0 \\ a_n \text{ è monotona decrescente} \\ \lim_{n \rightarrow +\infty} a_n = 0 \end{cases}$$

Allora converge

Serie di potenze

Raggio di convergenza

Sia $S(n) = \sum_{n=0}^{+\infty} a_n (x - x_0)^n$

- Se $\lim_{n \rightarrow +\infty} \sqrt[n]{|a_n|} = l$ o $\lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| = l$:
$$\begin{cases} \rho = \frac{1}{l} & \text{se } l \in (0, +\infty) \\ \rho = \infty & \text{se } l = 0 \\ \rho = 0 & \text{se } l = \infty \end{cases}$$

Calcolato ρ si ha che $S(n)$ converge in $(x_0 - \rho, x_0 + \rho)$

Derivata nel centro

Sia $S(x) = \sum_{n=0}^{+\infty} a_n (x - x_0)^n$

$$S^{(k)}(x_0) = k! a_k$$

Taylor

$$T_n(f(x); x_0) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

Funzioni elementari

- e^x

$$T_n(e^x) = \sum_{k=0}^n \frac{x^k}{k!}$$

- Seno/coseno

$$T_{2n+1}(\sin(x); 0) = \sum_{k=0}^n \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

$$T_{2n}(\cos(x); 0) = \sum_{k=0}^n \frac{(-1)^k x^{2k}}{(2k)!}$$

- $\frac{1}{1 \pm x}$

$$T_n\left(\frac{1}{1-x}; 0\right) = \sum_{k=0}^n x^k$$

$$T_n\left(\frac{1}{1+x}; 0\right) = \sum_{k=0}^n (-1)^k x^k$$

$$T_{2n}\left(\frac{1}{1+x^2}; 0\right) = \sum_{k=0}^n (-1)^k x^{2k}$$

- Logaritmo

$$T_n(\log(1+x); 0) = \sum_{k=1}^n (-1)^{k-1} \frac{x^k}{k}$$

- arctan

$$T_{2n+1}(\arctan(x); 0) = \sum_{k=0}^n \frac{(-1)^k x^{2k+1}}{2k+1}$$